It goes: (24, 18) —Ammu→ (18, 6) —Balu→ (12, 6) —Ammu→ (6, 6). Ammu wins again!

If they start with Ammu (18, 6), we get (18, 6) → (12, 6) → (6, 6) Balu wins!

Play the game with your friends and see for what pairs of numbers the first player (Ammu) wins, and when the second player wins.

Now we can notice something interesting! Begin with any pair of numbers. Can you say anything about the pair of numbers. Remember that we stop the process when both the numbers are same. It is the Highest Common Factor (HCF) of the two numbers we started with. So what we have seen here is an iterative process which leads us to the HCF of two given numbers. The HCF of a and b (here $a>b$) is the same as for a and $a-b$.

Example 1

Find the HCF of two numbers 16 and 28.

Solution

Now the HCF of 16,28	Now the HCF of (16 , 28-16)
16 = 2 x 2 x 2 x 2	16 = 2 x 2 x 2 x 2
28 = 2 x 2 x 7	12 = 2 x 2 x 3
HCF of (16, 28) = 2 x 2 = 4	HCF of (16, 12) = 2 x 2 = 4

Therefore HCF of (16, 28) = HCF of (16, 28-16).

Hence, HCF of two numbers a and b, $a > b$, is same as the HCF of a and $a - b$.

Euclidean algorithm

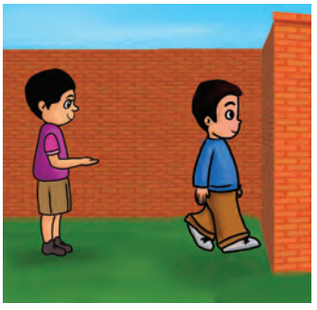
Let us take a number 12.

If we divide 12 by 7 then, we get quotient=1, remainder=5 and 12 can be written as 12 = (1 x 7) + 5.

If we divide 12 by 2 then, we get quotient=6, remainder=0 and 12 can be written as 12 = (2 x 6) + 0.

From this, we observe that, if a number 'a' is divided by some number 'b' then we get the quotient 'q' and remainder 'r', and 'a' can be written in a unique way as $a = (b \times q) + r$. That is, Dividend = (Divisor x Quotient) + Remainder. This is called the Euclidean algorithm.

5.4 Following and Devising Algorithms



Do you know the robot game? One child acts as a robot. Another child gives instructions to the child enacting as robot. The robot child should follow instructions. If the robot child stands at the wall by facing it, the instructor has to say, "move forward'. The robot child can only try to move forward, but can't. The robot child cannot say "it is not possible". This humorous activity shows that instructions are mechanically followed by robots. Unlike robots, human brain is capable of thinking and modifying algorithms based on situations.

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